Chapter 8C Edge colorings

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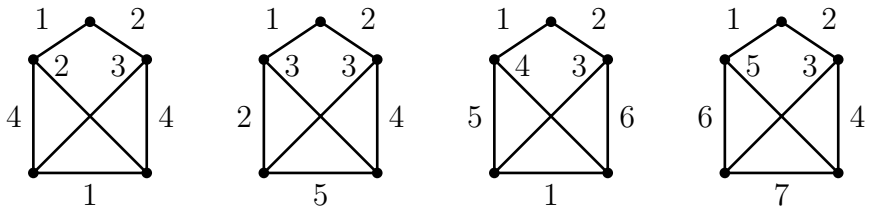
8C.1 Introduction and Basics

An obvious variation of vertex-coloring is edge-coloring where the edges are colored rather than vertices. The concept is useful to model many scheduling problems. It also arises in many circuit board problems where the wires connecting a device have to be of different color.

Throughout this chapter, graphs do not have loops (reason will be clear soon) but may have multiple edges.

Definition. *Any function $C : E(G) \to \{1, 2, \ldots, k\}$ is called a k-**edge-coloring** of G, if $C(x) \neq C(y)$ for any two adjacent edges x and y.*

Equivalently, a k-edge-coloring of G is a partition $(M_1, M_2, \ldots, M_k)$ of $E(G)$ such that each M_i is a matching in G. Throughout, we assume that the edges in M_i are colored i, $1 \leq i \leq k$. See Figures 8C.1 and 8C.2 for examples of edge-colorings.



(a) $k = 4$ (b) $k = 5$ (c) $k = 6$ (d) $k = 7$

Figure 8C.1: A k-edge-coloring of a graph G for $k = 4, 5, 6$ or 7. Notice that there is no k-edge-coloring of G if $k = 1, 2$ or 3.

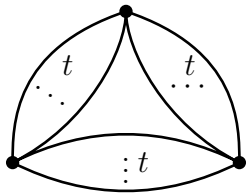


Figure 8C.2: An interesting graph for which only edge-coloring is the m-edge-coloring ($m = 3t$).

Definitions. *Let $C : E(G) \to \{1, 2, \ldots, k\}$ be an edge-coloring of G.*

- *An edge e colored β (that is, $C(e) = \beta$) is called a β-**edge.***
- *Let v be a vertex in G. If there is no β-edge incident with v, then we say that the color β is **missing** (or is **absent**) at v.*
- *If there is a β-edge incident with v, then we say that the color β **appears** (or is **present**) at v.*

Remarks.

- Every graph G admits a m-edge-coloring; color all the edges differently.
- Given k, a graph may not admit a k-edge-coloring.

The above remarks motivate the following concept.

Definition. *The minimum integer k such that G admits a k-edge-coloring is called the* ***edge-chromatic-number*** *of G. It is denoted by $\chi_1(G)$.*

- *Thus, $\chi_1(G)$ is the minimum number of colors required to color the edges of G such that no two adjacent edges receive the same color.*
- *The edge-chromatic-number is also called the* ***chromatic index****.*

Remarks.

- $\chi_1(G) \geq \Delta(G)$, since in any edge-coloring, the colors appearing at any vertex are all distinct. More generally, if C is an edge-coloring and $C(v)$ denotes the number of colors appearing at v, then $C(v) = deg(v)$.
- A natural question is to ask how much $\chi_1(G)$ is bigger than $\Delta(G)$. A remarkable theorem, independently proved by V.G. Vizing (1964) and R. P. Gupta (1966) states that $\chi_1(G) \leq \Delta(G) + 1$, for any simple graph G. We shall prove this theorem soon.

The following table shows the edge-chromatic-number of some elementary graphs. Proofs for the last two entries require some efforts.

G	P_n	C_n, n even	C_n, n odd	Petersen graph	$K_{r,s}$	K_n, n even	K_n, n odd
$\chi_1(G)$	2	2	3	4	$\max\{r, s\}$	$n-1$	n

Table 8C.1: A table of edge-chromatic numbers, where $n \geq 3$.

8C.2 Gupta-Vizing theorem

Theorem 8C.1 (V. G. Vizing (1964), R. P. Gupta (1966))**.** *For any simple graph G,* $\Delta(G) \leq \chi_1(G) \leq \Delta(G) + 1$.

Proof. (Indirect Induction/Contradiction method) As observed before, the lower bound is obvious. So, we have to only prove the upper bound. Assume the contrary

and let G be a graph having the minimum number of edges among all the graphs with $\chi_1 \geq \Delta + 2$. So, $\chi_1(G) \geq \Delta(G) + 2$ but $\chi_1(G - e) \leq \Delta(G - e) + 1 \leq \Delta(G) + 1$, for every edge e.

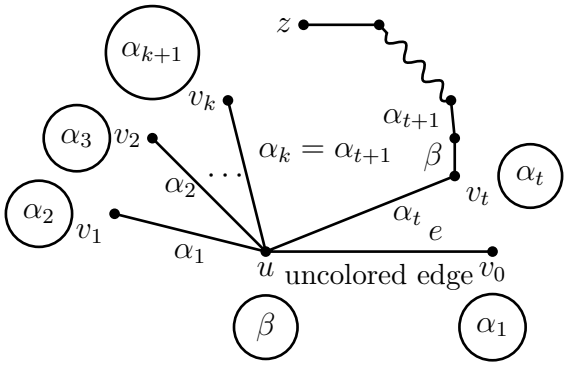


Figure 8C.3: Missing colors are shown inside the circles.

Let C be a $(\Delta+1)$-edge-coloring of $G-e$, where e is (u, v_0). Before proceeding further, we remark that:

- At any vertex x at least one color is absent, since there are at most Δ edges incident with x and we have $\Delta + 1$ colors.
- If there exists a $(\Delta + 1)$-edge-coloring C' of $G - e$ such that some color β is absent at u and at v_0, then e can be colored with β to get a $(\Delta+1)$-edge-coloring of G, which provides a contradiction to our choice of G.

Using a recoloring technique, we show that it is always possible to get an edge-coloring C' as described above.

Let α_1 be a color missing at v_0. There is an α_1-edge incident with u; else, by coloring (u, v_0) with α_1 we get a $(\Delta + 1)$-edge-coloring of G. Let (u_1, v_1) be the edge colored α_1; see Figure 8C.3.

Let α_2 be a color missing at v_1. There is an α_2-edge incident with u; else, by recoloring (u, v_1) with α_2 and (u, v_0) with α_1 we get a $(\Delta+1)$-edge-coloring of G. So, there must exist an edge say (u, v_2) colored α_2.

Let α_3 be a color missing at v_2. There is an α_3-edge incident with u; else, by recoloring the edges (u, v_2), (u, v_1) and (u, v_0) with colors α_3, α_2 and α_1, respectively we get a $(\Delta + 1)$-edge-coloring of G.

We continue this process to generate:

- a sequence of distinct vertices $v_0, v_1, v_2, \ldots$ incident with u, and
- a sequence of colors $\alpha_1, \alpha_2, \alpha_3, \ldots$ such that the edges $(u, v_1), (u, v_2), \ldots$ are colored $\alpha_1, \alpha_2, \ldots$ respectively, and α_{i+1} is the color missing at $v_i, i \geq 0$.

Since the degree of u is finite, the sequence $v_0, v_1, v_2, \ldots$ is finite. Therefore, there exists some v_t such that its missing color $\alpha_{t+1} \in \{\alpha_1, \ldots, \alpha_t\}$. Let t be the smallest integer such that $\alpha_{t+1} = \alpha_k$, where $\alpha_k \in \{\alpha_1, \ldots, \alpha_t\}$. This is equivalent to saying that there exists an integer k such that (u, v_k) is colored $\alpha_{t+1} (= \alpha_k)$, where $1 \leq k < t$.

Let β be a color missing at u but appearing at v_0. Such a color exists; else we can color (u, v_0) with β to get a $(\Delta + 1)$-edge-coloring of G.

Claim: β appears at every v_i, $0 \leq i \leq t$.

To prove the claim, assume the contrary and let p be the smallest integer such that β is absent at v_p $(1 \leq p \leq t)$. Recolor (u, v_p) with β and edges (u, v_{p-1}), $(u, v_{p-2}), \ldots, (u, v_0)$ with colors $\alpha_p, \alpha_{p-1}, \ldots, \alpha_1$ to get a $(\Delta + 1)$-edge-coloring of G. So the claim holds.

We define a path P as follows:

P is a maximal path with origin v_t and its edges colored alternately β and α_{t+1}. Suppose P terminates at a vertex z. At the outset, observe that the first edge and the last edge of P are colored β; see the claim above. Also $v_t \neq v_k$.

We consider three different cases depending on the location of z, and in each case describe a technique to recolor the edges $G - e$ which leads to a $(\Delta + 1)$-edge-

coloring of G.

Case 1: $z \in V(G) - \{v_0, v_1, \ldots, v_t, u\}$.

Interchange the colors β and α_{t+1} of P. After this interchange, there is no β-edge incident with v_t. Recolor (u, v_t) with β, and edges $(u, v_{t-1}), \ldots, (u, v_0)$ with colors $\alpha_t, \ldots, \alpha_1$, respectively. This yields a $(\Delta + 1)$-edge-coloring of G.

Case 2: $z = v_j$, for some j, $0 \leq j \leq t - 1$.

Interchange the colors β and α_{t+1} of P. After this interchange, there is no β-edge incident with v_j. Recolor (u, v_j) with β, and recolor the edges $(u, v_{j-1}), \ldots, (u, v_0)$ with colors $\alpha_j, \ldots, \alpha_1$, respectively. Again we obtain a $(\Delta + 1)$-edge-coloring of G.

Case 3: $z = u$. In this case, the last edge of P is (v_k, u). Interchange the colors β and α_{t+1} of P. After this interchange, (v_k, u) is colored β. Recolor the edges $(u, v_{k-1}), \ldots, (u, v_0)$ with colors $\alpha_k, \ldots, \alpha_1$, respectively. This yields a $(\Delta + 1)$-edge-coloring of G. □

The above theorem does not hold for graphs with multiple edges. See Figure 8C.4.

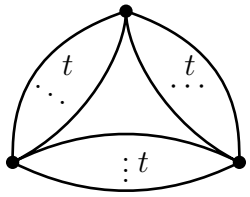


Figure 8C.4: $\Delta(G) = 2t, \chi_1(G) = 3t = \Delta(G) + t$.

The following is the generalization of the above theorem for multigraphs.

Theorem 8C.2 (V. G. Vizing (1965), R. P. Gupta (1966))**.** *For any loopless graph G,*

$$\chi_1(G) \leq \Delta_1(G) + \mu(G),$$

where $\mu(G)$ denotes the maximum number of edges joining a pair of vertices in G.

8C.3 Class-1 and Class-2 graphs

Theorem 8C.1 divides the class of simple graphs into two classes.

Definition. *A simple graph G is said to be of* ***Class-1*** *if $\chi_1(G) = \Delta(G)$ and it is said to be of* ***Class-2*** *if $\chi_1(G) = \Delta(G) + 1$.*

The problem of finding necessary and sufficient conditions for a graph to be of Class-1 (or equivalently Class-2) is a hard unsolved problem. It is called the ***Classification problem***. In the following, we state and prove a few necessary conditions and a few sufficient conditions for a graph to be of Class-1 or Class-2. An alert reader may have noticed that we made similar remarks on the characterization of Hamilton graphs (Chapter 6). It is known that Class-2 graphs are rare. More precisely, it has been shown that if we randomly pick a graph G on n vertices, then the probability that G is Class-1, tends to 1 as $n \to \infty$. For example, among the 143 non-isomorphic simple graphs on at most six vertices only 8 belong to Class-2. See Figure 8C.5

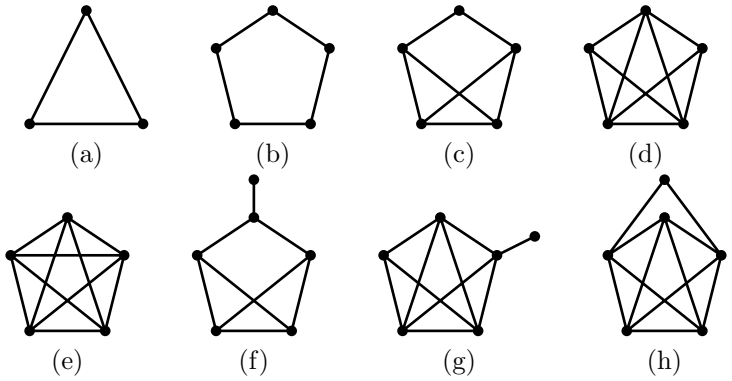


Figure 8C.5: All the Class-2 graphs with at most six vertices.

- **Edge-coloring of bipartite graphs**

The best known theorem on Class-1 graphs is the following.

Theorem 8C.3 (König, 1916)**.** *For any bipartite graph G,*

$$\chi_1(G) = \Delta(G).$$

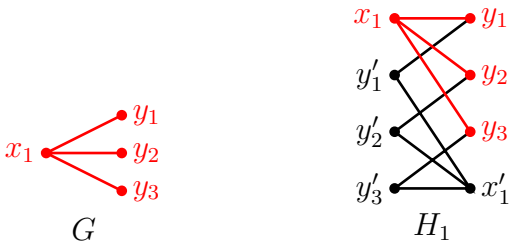
Proof. At the outset, observe that it is enough to prove the theorem for connected graphs, since $\chi_1(G) = \max\{\chi_1(D) : D \text{ is a component of } G\}$. We first construct a $\Delta(G)$-regular bipartite graph H containing G as an induced subgraph. Let $[X, Y]$ be the bipartition of $V(G)$, where $X = \{x_1, x_2, \ldots, x_r\}$ and $Y = \{y_1, y_2, \ldots, y_s\}$. Let $G^1[X^1, Y^1]$ be an isomorphic copy of G, where $X^1 = \{x'_1, x'_2, \ldots, x'_r\}$ and $Y^1 = \{y'_1, y'_2, \ldots, y'_s\}$; see Figure 8C.6a. Construct the bipartite graph $H_1[X \cup Y^1, X^1 \cup Y]$ by adding the following edges:

- Join x_i and x'_i $(1 \le i \le r)$, if $\deg_G(x_i) < \Delta(G)$.

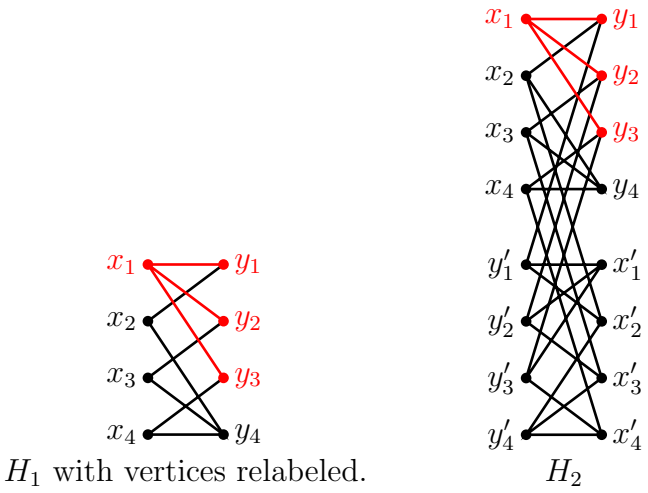
- Join y_i and y'_i $(1 \le i \le s)$, if $\deg_G(y_i) < \Delta(G)$.

Then $\Delta(H_1) = \Delta(G)$ and $\delta(H_1) = \delta(G) + 1$. If $\delta(G) + 1 = \Delta(G)$, then H_1 is a $\Delta(G)$-regular bipartite graph with G as an induced subgraph. If $\delta(G)+1 < \Delta(G)$, we repeat the above process with H_1 to construct a bipartite H_2 which contains H_1 (and hence G) as an induced subgraph such that $\Delta(H_2) = \Delta(G)$ and $\delta(H_2) = \delta(H_1)+1 = \delta(G)+2$. If $\delta(G)+2 = \Delta(G)$, then H_2 is a required graph. We can continue the process enough number of times and construct a sequence of bipartite graphs $H_1, H_2, \ldots, H_t$ such that $G \sqsubseteq H_i \sqsubseteq H_{i+1}$, $\delta(H_{i+1}) = \delta(H_i) + 1$ $(1 \le i \le t-1)$, $\Delta(H_i) = \Delta(G)$ $(1 \le i \le t)$ and $\delta(H_t) = \Delta(G)$. Then H_t is a required bipartite graph.

By Corollary to Hall's Theorem 7.3, $E(H_t)$ can be partitioned into $\Delta(G)$ perfect matchings $E_1, E_2, \ldots, E_{\Delta(G)}$. Therefore, $\chi_1(H_t) \le \Delta(G)$. Since $G \sqsubseteq H_t$, we have $\chi_1(G) \le \chi_1(H_t) \le \Delta(G) \le \chi_1(G)$ and so $\chi_1(G) = \Delta(G)$. □



(a) $G \to H_1$



(b) $H_1 \to H_2$

Figure 8C.6: Construction of a 3-regular bipartite graph which contains G as an induced subgraph.

8C.4 Scheduling problem and equitable edge-coloring

An example of a scheduling problem: A conference is planned with five invited speakers S_1, S_2, S_3, S_4, S_5 to handle six short courses $C_1, C_2, C_3, C_4, C_5, C_6$. For the convenience of the participants, on any day a speaker speaks at most once and a course is scheduled at most once.

The following matrix $M = [m_{ij}]_{5\times 6}$ describes the lecture requirements, where m_{ij} is the number of times speaker S_i speaks on the course C_j.

	C_1	C_2	C_3	C_4	C_5	C_6
S_1	0	0	1	0	3	0
S_2	0	2	0	1	0	1
S_3	2	0	0	0	2	0
S_4	0	1	0	2	0	2
S_5	2	0	2	0	0	0

(a) Find the smallest number of days needed to conduct the conference.

(b) Is it possible to schedule the time table where there are only four conference halls available

Graph theory model and a solution.

Associate the bipartite graph $G[S, C]$ shown in Figure 8C.7 with the given matrix M, where S_i and C_j are joined with m_{ij} edges.

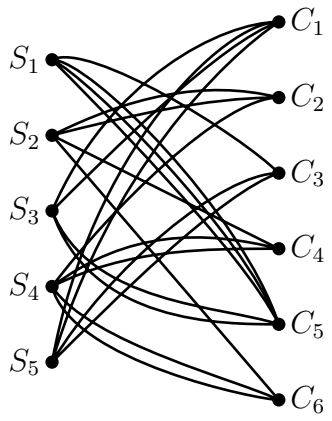


Figure 8C.7: The bipartite graph G associated with M.

Because of the restrictions on the scheduling of speakers and courses, the problem (a), is to partition the edge set of G into subsets $M_1, M_2, \ldots, M_t$ such that

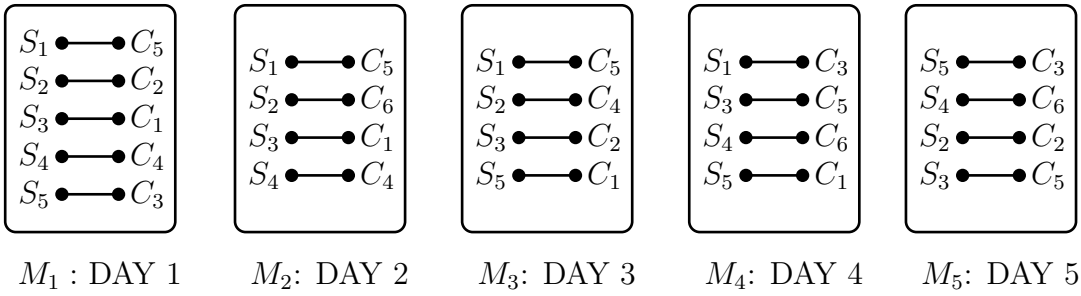


Figure 8C.8: Five disjoint matchings of G and the resultant schedule.

each M_i is a matching and t is minimum possible. Obviously, M_i is the schedule for the i-th day. Hence, in the terminology of edge-coloring, we have to find $\chi_1(G)$. Thanks to König's Theorem 8C.3, we immediately have that $\chi_1(G) = \Delta(G) = 5$. So, the answer to problem (a) is that five days are required to conduct the conference. An actual schedule is shown in Figure 8C.8. And a conference time table derived using this schedule is shown in Table 8C.2.

	Course 1	Course 2	Course 3	Course 4	Course 5	Course 6
Day 1	S_3	S_2	S_5	S_4	S_1	–
Day 2	S_3	–	–	S_4	S_1	S_2
Day 3	S_5	S_4	–	S_2	S_1	–
Day 4	S_5	–	S_1	–	S_3	S_4
Day 5	–	S_2	S_5	–	S_3	S_4

Table 8C.2: Conference time table derived using the schedule given in Figure 8C.8.

We next solve problem (b).

Since there are twenty one edges and there are 5 sets of matchings, at least one matching, say M_1 contains 5 or more edges, by the Pigeon-hole-principle. Since an edge in M_i indicates the requirement of a conference hall on i-th day, 5 conference halls are necessarily required. So, the answer to problem (b) is NO.

If there are only four conference halls available we require six days to conduct the conference. It is easy to modify the table so that every day we require at most four conference halls. For example, we can shift the edge (S_5, C_3) which appears in M_1 to a new matching M_6. But the resultant time table is obviously unbalanced: only one talk on sixth day. So to have a balance, we would like to have a 6-edge-coloring of G such that $||M_i| - |M_j|| \leq 1$, $1 \leq i, j \leq 6$, that is, the number of talks are "nearly equal" on all the days. Next theorem asserts that it is always possible to schedule the time table in a balanced manner.

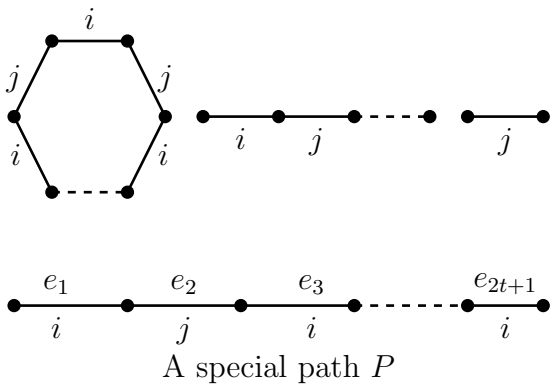
Definition. *A k-edge-coloring $C = (M_1, M_2, \ldots, M_k)$ is said to be a* ***equitable k-edge-coloring*** *if $||M_i| - |M_j|| \leq 1$, for every $i, j \in \{1, 2, \ldots, k\}$.*

Theorem 8C.4. *If there exists a k-edge-coloring of G, then there exists an equitable k-edge-coloring of G.*

Proof. Let $C = (M_1, M_2, \ldots, M_k)$ be a k-edge-coloring of G. Suppose there exist i and j such that $|M_i| \geq |M_j| + 2$.

We describe a procedure to get matchings M_i' and M_j' such that $|M_i'| = |M_i| - 1$, $|M_j'| = |M_j| + 1$ and $M_i \cup M_j = M_i' \cup M_j'$. By repeating such a procedure enough times, we can get an equitable k-edge-coloring of G.

Consider the subgraph H of G induced by the set of edges $M_i \cup M_j$. $\Delta(H) \leq 2$, since there can be at most one i-edge and at most one j-edge incident with any vertex in H. So, every component of H is either a cycle or a path whose edges are alternately colored i and j. Therefore, any cycle component is an even cycle; see Figure 8C.9. We conclude that there exists a path component P in H whose first edge and last edge are colored i, since $|M_i| \geq |M_j| + 2$.



A special path P

Figure 8C.9: Components of H.

Define

$$\begin{aligned} M_i' &= M_i - \{e_1, e_3, \ldots, e_{2t+1}\} \cup \{e_2, e_4, \ldots, e_{2t}\}, \\ M_j' &= M_j - \{e_2, e_4, \ldots, e_{2t}\} \cup \{e_1, e_3, \ldots, e_{2t+1}\}. \end{aligned}$$

That is, interchange the colors of P (and retain the colors of edges not in P). Then $M_i \cup M_j = M_i' \cup M_j'$, $|M_i'| = |M_i| - 1$, and $|M_j'| = |M_j| + 1$. □